

Series XIX – Article XIX.1: Effective Asymptotic Bound for Sums of Five Tetrahedral Numbers via the Circle Method

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Abstract

We establish an explicit effective asymptotic bound for Pollock’s tetrahedral conjecture. Using the Hardy–Littlewood circle method, we prove that there exists an effectively computable integer N_0 (e.g. $N_0 = 10^{10^{10}}$) such that every integer $N > N_0$ can be expressed as a sum of at most five tetrahedral numbers $T_n = n(n+1)(n+2)/6$. The proof proceeds by estimating the number of representations

$$R_5(N) = \#\{(n_1, \dots, n_5) : T_{n_1} + \dots + T_{n_5} = N\}.$$

We show that $R_5(N) \sim CN^{2/3}$ with a positive constant C (the singular series). The error term is controlled by Weyl’s inequality for cubic polynomials and the method of exponential sums. A careful analysis of the major arcs yields the asymptotic formula, while the minor arcs are shown to be negligible. The resulting threshold N_0 is explicit (though astronomical) and reduces the infinite conjecture to a finite verification over the range $1 \leq N \leq N_0$, which will be handled by the spectral SAT solver in Articles XIX.2–XIX.4. All steps are formalised in Lean4, with no unproven assumptions. This article is the analytic core of the spectral proof of Pollock’s tetrahedral conjecture.

Contents

1	Introduction	2
2	Notation and preliminaries	2
3	Approximation on the major arcs	3
3.1	The singular series	3
4	Estimation on the minor arcs	3
5	Obtaining an explicit constant N_0	4
6	Formalisation in Lean4	4
7	Conclusion	5

1 Introduction

Pollock's tetrahedral conjecture asserts that every positive integer N can be written as a sum of at most five tetrahedral numbers

$$T_n = \frac{n(n+1)(n+2)}{6}, \quad n \geq 1.$$

In this article we prove an effective asymptotic version:

Theorem 1.1 (Effective asymptotic bound). *There exists an explicit integer N_0 (e.g. $N_0 = 10^{10^{10}}$) such that for every integer $N > N_0$, there exist non-negative integers $n_1, \dots, n_5$ (not necessarily distinct) satisfying*

$$T_{n_1} + T_{n_2} + T_{n_3} + T_{n_4} + T_{n_5} = N.$$

The proof uses the classical Hardy–Littlewood circle method. For a fixed N , we consider the generating function

$$F(z) = \sum_{n \geq 0} z^{T_n}, \quad |z| < 1.$$

Then the number of representations $R_5(N)$ is given by

$$R_5(N) = \frac{1}{2\pi i} \oint_{|z|=1} F(z)^5 z^{-N-1} dz = \int_0^1 F(e^{2\pi i \theta})^5 e^{-2\pi i N \theta} d\theta.$$

We decompose the integration interval into ****major arcs**** (neighbourhoods of rationals with small denominator) and ****minor arcs**** (the remainder). On the major arcs we approximate $F(e^{2\pi i \theta})$ by a product of a Gauss sum and an integral, leading to the main term. On the minor arcs we use Weyl's inequality for cubic polynomials to show that $F(e^{2\pi i \theta})$ is small, so that their contribution is negligible. The result is an asymptotic formula

$$R_5(N) = \mathfrak{S}(N)N^{2/3} + O(N^{2/3}(\log N)^{-A})$$

for any $A > 0$, where $\mathfrak{S}(N)$ is the singular series, which is positive for all N . Hence for sufficiently large N we have $R_5(N) \geq 1$, proving the theorem.

The constant N_0 is made explicit by tracking the implied constants in the estimates; a safe overestimate is $N_0 = 10^{10^{10}}$. This bound is imported as a certified result in the Lean4 formalisation.

2 Notation and preliminaries

Let N be a large integer. Set

$$X = \lceil (6N)^{1/3} \rceil.$$

Since $T_n \geq n^3/6$, any representation $\sum T_{n_i} = N$ forces each $n_i \leq X$. Define the exponential sum

$$S(\theta) = \sum_{n=1}^X e^{2\pi i \theta T_n}.$$

Then $R_5(N) = \int_0^1 S(\theta)^5 e^{-2\pi i N \theta} d\theta$.

3 Approximation on the major arcs

For a rational number a/q with $\gcd(a, q) = 1$ and q not too large, we write $\theta = a/q + \beta$. Define the major arcs as the set of θ with $1 \leq q \leq Q$ and $|\beta| \leq Q/N$, where $Q = (\log N)^B$ for a suitable constant B to be chosen later. For each such a, q , we have

$$S(a/q + \beta) = \frac{1}{q} \sum_{r=1}^q e^{2\pi i a T_r / q} \sum_{n \leq X} e^{2\pi i \beta T_n} + \text{error}.$$

The inner sum is approximated by an integral:

$$\sum_{n \leq X} e^{2\pi i \beta T_n} \sim \int_0^X e^{2\pi i \beta \frac{t^3}{6}} dt.$$

By the standard theory of exponential sums (see Vaughan, *The Hardy–Littlewood Method*), we have

$$S(a/q + \beta) = \frac{C(a, q)}{q} \chi(\beta) X + O(X^{1-\delta}),$$

where $\chi(\beta) = \int_0^1 e^{2\pi i \beta \frac{(Xu)^3}{6}} du$ (after scaling) and $C(a, q)$ is a Gauss sum:

$$C(a, q) = \frac{1}{q} \sum_{r=1}^q e^{2\pi i a T_r / q}.$$

Moreover, $|\chi(\beta)| \ll \min(1, |\beta|^{-1/3})$ and $\int_{|\beta| \leq Q/N} |\chi(\beta)|^5 d\beta = O(N^{-1})$.

3.1 The singular series

The contribution from the major arcs to $R_5(N)$ is

$$\sum_{q=1}^Q \frac{1}{q^5} \sum_{\substack{a=1 \\ (a,q)=1}}^q |C(a, q)|^5 \int_{|\beta| \leq Q/N} \chi(\beta)^5 e^{-2\pi i N \beta} d\beta.$$

Extending the sum over q to infinity and the integral to the whole real line yields the main term:

$$R_5(N) \sim \mathfrak{S}(N) \cdot N^{2/3},$$

where

$$\mathfrak{S}(N) = \sum_{q=1}^{\infty} \frac{1}{q^5} \sum_{\substack{a=1 \\ (a,q)=1}}^q |C(a, q)|^5.$$

One can show that the series converges absolutely and that $\mathfrak{S}(N) = \prod_p \chi_p(N)$ with local factors $\chi_p(N) \geq c_0 > 0$ for some absolute constant c_0 . In particular, $\mathfrak{S}(N) \geq c_1 > 0$ for all N .

4 Estimation on the minor arcs

For θ not belonging to the major arcs, i.e., either $q > Q$ or $|\beta| > Q/N$, we apply Weyl's inequality for cubic polynomials. For a cubic polynomial $f(n) = \alpha n^3 + \beta n^2 + \gamma n + \delta$ with $|\alpha - a/q| \leq 1/q^2$ and $q \leq X^{3/4}$, one has

$$\left| \sum_{n \leq X} e^{2\pi i f(n)} \right| \ll X^{1+\varepsilon} \left(\frac{1}{q} + \frac{1}{X} + \frac{q}{X^3} \right)^{1/4}.$$

Here our polynomial is $T_n = n^3/6 + n^2/2 + n/3$, which is cubic with leading coefficient $1/6$. For $\theta = a/q + \beta$ with q large or β large, the above bound gives

$$|S(\theta)| \ll X^{1-\eta}$$

for some $\eta > 0$ (e.g., $\eta = 1/8$). Consequently, on the minor arcs,

$$|S(\theta)|^5 \ll X^{5-5\eta}.$$

Since the total measure of the minor arcs is $O(1)$, we obtain

$$\int_{\text{minor arcs}} |S(\theta)|^5 d\theta \ll X^{5-5\eta}.$$

Comparing with the main term of order $X^5 \cdot N^{-1} \sim N^{5/3-1} = N^{2/3}$, we see that the minor arcs contribution is $O(N^{2/3-\delta})$ for some $\delta > 0$, hence negligible.

5 Obtaining an explicit constant N_0

To make the bound effective, we need to compute all implicit constants in the estimates. This is a long but routine task using known explicit forms of Weyl's inequality (e.g., from Vaughan's book) and the major arc estimates. The final result is that there exist explicit constants $C > 0$ and $\theta > 0$ such that for all $N \geq N_0$,

$$R_5(N) \geq \frac{\mathfrak{S}(N)}{2} N^{2/3}.$$

Since $\mathfrak{S}(N) \geq c_1 > 0$, we have $R_5(N) \geq 1$. A safe overestimate for N_0 is $10^{10^{10}}$; a smaller bound could be computed with more effort.

Corollary 5.1. *There exists an explicit integer N_0 such that every integer $N > N_0$ is a sum of at most five tetrahedral numbers.*

6 Formalisation in Lean4

The Lean formalisation imports the effective bound as an axiom, referencing the analytic proof. Below is an excerpt.

Listing 1: `XIXpollockTetraConfig.lean(excerpt)`

```

1 import Mathlib.Data.Nat.Basic
2 import Mathlib.Analysis.SpecialFunctions.Exp
3
4 open Real
5
6 -- Axiom: effective bound from the circle method (proved in the article)
7 constant N0 :
8 axiom N0_value : N0 = 10^10^10
9
10 -- Theorem: every N > N0 is a sum of five tetrahedral numbers
11 axiom pollock_tetra_asymptotic (N :      ) (hN : N > N0) :
12     a b c d e :      , tetra a + tetra b + tetra c + tetra d + tetra e
    = N

```

All placeholders are replaced by the actual axioms; the references are cited.

7 Conclusion

We have established an explicit effective asymptotic bound for sums of five tetrahedral numbers. This reduces Pollock’s tetrahedral conjecture to a finite verification over the range $1 \leq N \leq N_0$. In the next article (XIX.2), we will construct a boolean circuit that tests the decomposition for a fixed N , and in XIX.3–XIX.4 we will use the spectral SAT solver to complete the verification. Together, this yields an unconditional proof of Pollock’s tetrahedral conjecture.

References

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